

Akshat Sharma

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EDUCATION

Maharaja Agrasen Institute of Technology

B.Tech. in Electronics and Communication Engineering | CGPA: 8.46/10

Rohini, Delhi

2023–2027

EXPERIENCE

Embedded Systems Engineering Intern @ Rebhu Computing

- Developed FPGA-based matrix multiplication accelerators using the Artix-7 100T to optimize the performance of AI models.
- Implemented **Bluetooth LE PAwR (Periodic Advertising with Responses)** for scalable, bidirectional communication between ESL (Electronic Shelf Label) tags and access points.
- Designed and developed low-level **device drivers** for SPI-based display modules, enabling efficient display updates and reduced power consumption.
- Worked on board bring-up, hardware validation, and peripheral integration for embedded platforms.

Open Source Hardware Intern @ FOSSEE, IIT Bombay

- Contributed to the integration of **ESP32 emulation** into FOSSEE's Yaksh automated evaluation platform.
- Worked with QEMU-based emulation environments for ESP32 firmware execution and validation.
- Developed and tested workflows supporting both **MicroPython** and **ESP-IDF** applications.
- Assisted in creating reproducible embedded software test environments for large-scale student evaluation.

Robotics Intern @ Rebhu Computing

- Developed an autonomous **UGV platform** using ArduRover and NVIDIA Jetson Orin.
- Created a custom **ICP-based module** for odometry estimation from point-cloud data.
- Fused wheel-encoder data received via CAN bus with IMU data for enhanced localization.
- Worked on ArduPilot–Gazebo SITL simulations.
- Set up RTSP camera video-capture pipelines using GStreamer.

PROJECTS

e-Yantra Robotics Competition 2025 – Verilog

[Project Link](#)

- Built a **maze-solving robot** using the DE0-Nano FPGA board.
- Implemented **PWM generation**, distance measurement using ultrasonic sensors, UART receiver and transmitter, and a DHT11 humidity and temperature sensor interface.
- Designed and implemented a RISC-V CPU in Verilog supporting all 40 RV32I base instructions.
- Simulated, synthesized, and verified** the design using Intel Quartus Prime and ModelSim.

WS2812B DMA-Based Driver – STM32

[Project Link](#)

- Implemented a high-speed driver for WS2812B addressable LEDs on an STM32F103C6T6 by configuring **hardware timers** for precise PWM signal generation.
- Optimized CPU utilization by offloading data transmission to the **DMA (Direct Memory Access)** controller, ensuring accurate timing for the single-wire serial protocol.
- Verified signals and timing on an oscilloscope.

CHIP-8 Interpreter on FPGA – Verilog | DE0-Nano

[Project Link](#)

- Built a **CHIP-8 interpreter in Verilog** to run classic CHIP-8 games directly on the DE0-Nano FPGA.
- Integrated **VGA display, PS/2 keyboard input, and audio output** to create a complete standalone gaming system.
- Implemented support for loading and running different CHIP-8 ROMs, with game execution and I/O handled entirely on the FPGA.
- Simulated the design using Verilator** and tested the complete system on physical FPGA hardware.

SKILLS

Languages: C/C++ | Python | Verilog

Technologies: STM32 | ESP32 | BLE | Embedded Linux | ROS 2 | Docker | Git | QEMU | KiCad | Fusion 360

Skills: Embedded Systems | Bare-Metal Firmware | Device Drivers | FPGA Design | RISC-V | PCB Design | Robotics